WEEK 6 FINAL ASSIGNMENT

ARTIFICIAL INTELLIGENCE

Student Performance Prediction Using Machine Learning

Assignment Type: Final Project
Duration: 1 Week (7 Days) **Total Marks:** 100

Prepared as an educational demonstration of the end-to-end supervised machine-learning workflow.

Dataset notice: This project uses a synthetic dataset generated for learning and reproducibility. It does not contain real student records and the results are not evidence of real-world predictive performance.

1. Introduction

Student performance can be influenced by multiple factors, including study time, attendance, previous achievement, sleep, tutoring, and access to learning resources. Machine learning can model patterns in historical data and estimate a numerical outcome. This project demonstrates a complete regression workflow: problem definition, dataset inspection, cleaning, exploratory data analysis, preprocessing, training, prediction, evaluation, and conclusion.

2. Problem Statement and Objectives

Problem statement: Build a model that estimates a student's final score from selected study-habit and background features.

Objectives: (1) inspect and clean tabular data; (2) visualize patterns; (3) preprocess numerical and categorical variables; (4) train a Random Forest regressor; (5) evaluate test-set predictions with MAE, MSE, RMSE, and R^2.

Why ML is suitable: The task has a numerical target and several potentially interacting predictors. A supervised regression model learns relationships from labelled examples and produces continuous score estimates.

Expected outcome/application: Demonstrate a reproducible model and identify patterns worth further study. In a real education setting, any use would require validated data, consent, fairness assessment, human oversight, and safeguards against using predictions to limit student opportunities.

3. Dataset Description

The included CSV contains 1,012 rows before cleaning (1,000 generated examples plus 12 intentional duplicate rows) and 9 columns. It is synthetic and reproducibly generated using NumPy with a fixed random seed. Features include hours_studied, attendance_pct, previous_score, sleep_hours, tutoring, parental_education, extracurricular, and internet_access. The target is final_score (0–100).

Source: Generated for this assignment; no external personal data source is used. This should be disclosed in any submission or presentation.

Item	Before cleaning	After cleaning
Rows	1012	1000
Columns	9	9
Missing cells	60	0
Duplicate rows	12	0

Cleaning procedure: remove exact duplicate records; fill missing numeric values with the median and missing categorical values with the mode. The modelling pipeline repeats imputing and encoding within the training workflow to keep preprocessing reproducible. Numeric features are standardized and categorical features are one-hot encoded.

4. Exploratory Data Analysis

The following visualizations are generated from the cleaned dataset. They describe this synthetic sample only; correlation and group differences do not establish causation.

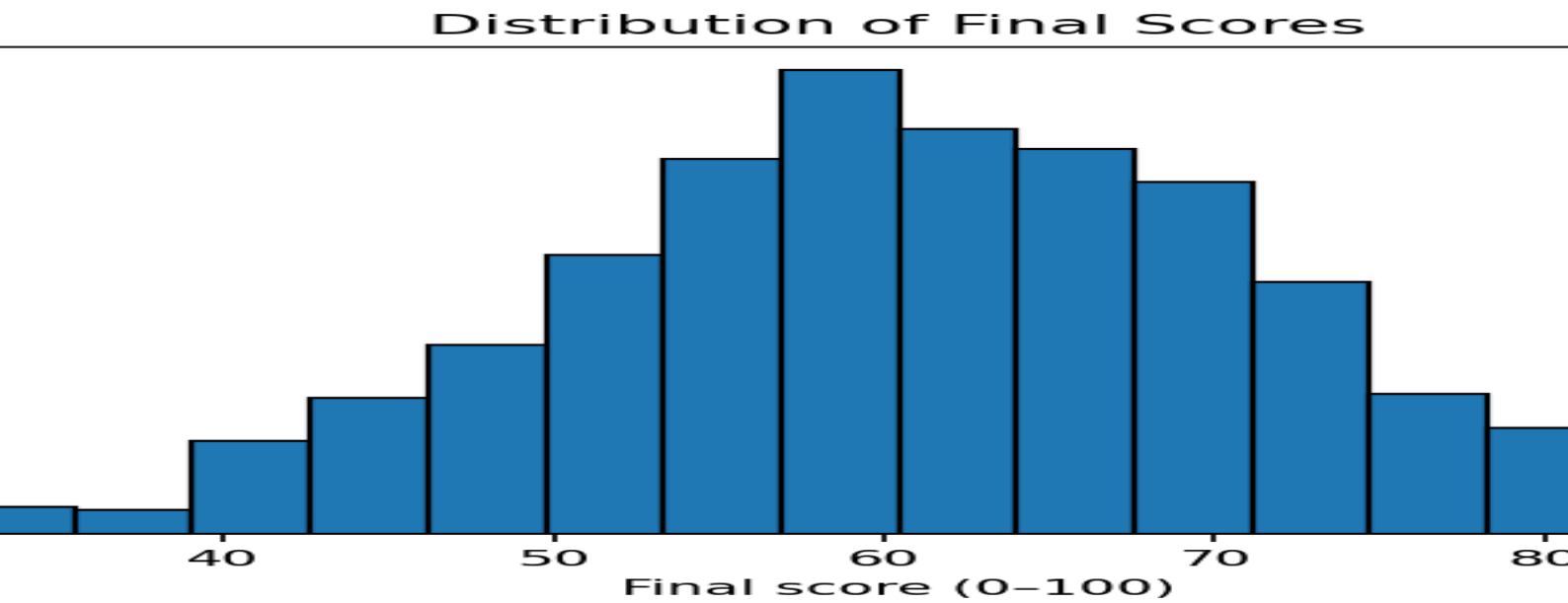


Figure 1 — Final-score distribution. Shows the spread and concentration of scores, helping identify the general range and any skew.

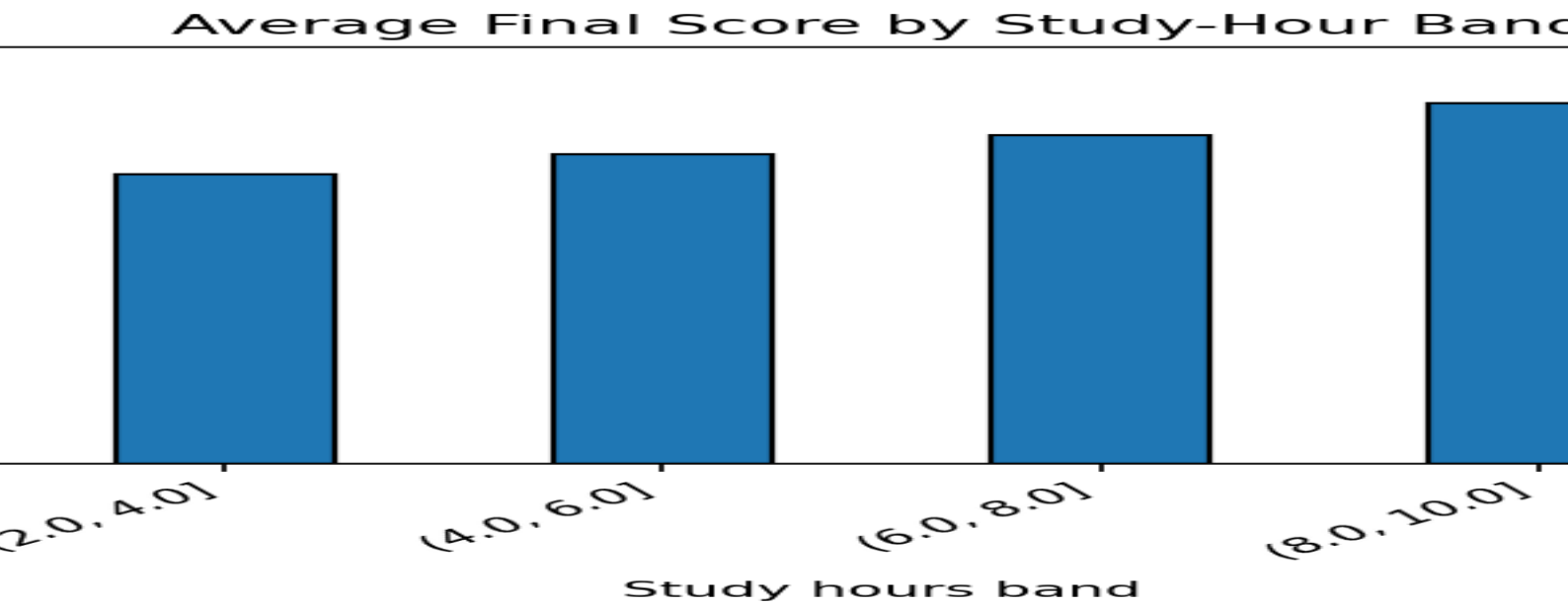


Figure 2 — Average score by study-hour band. Compares average scores across study-time groups; differences are descriptive and may reflect other factors.

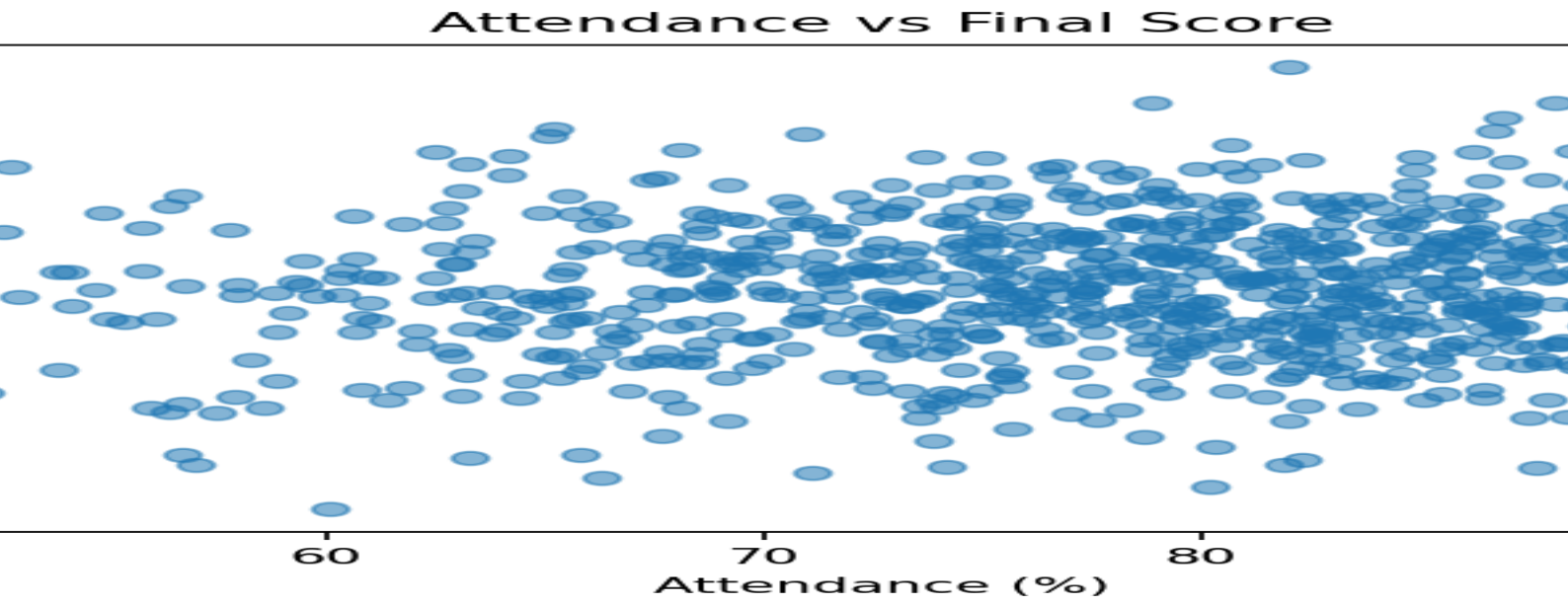


Figure 3 — Attendance versus final score. The scatter plot helps assess whether attendance and scores show an association and how much variation exists at similar attendance levels.

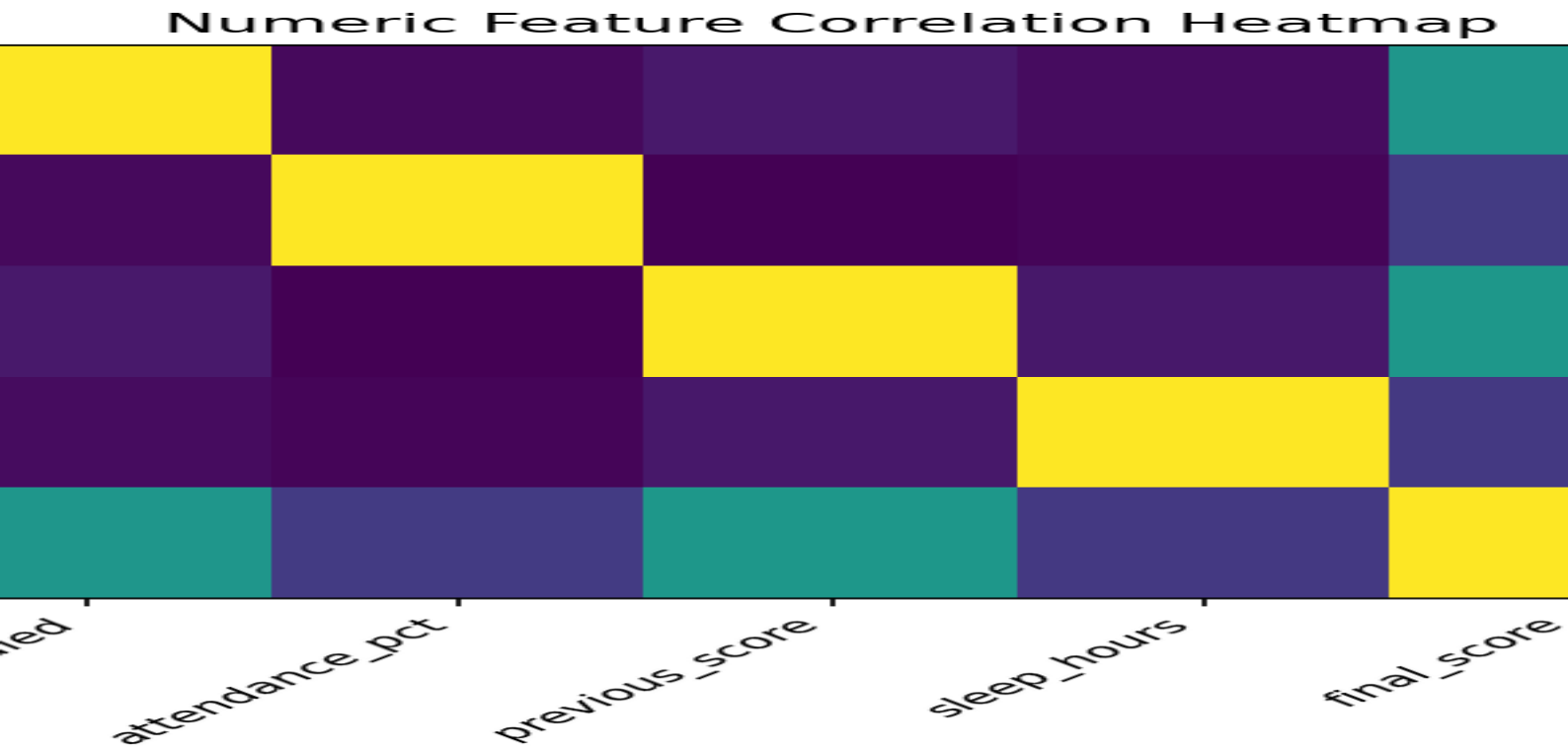


Figure 4 — Correlation heatmap. Displays pairwise linear correlations between numerical columns. Values near +1 indicate positive association, near −1 negative association, and near 0 weak linear association.

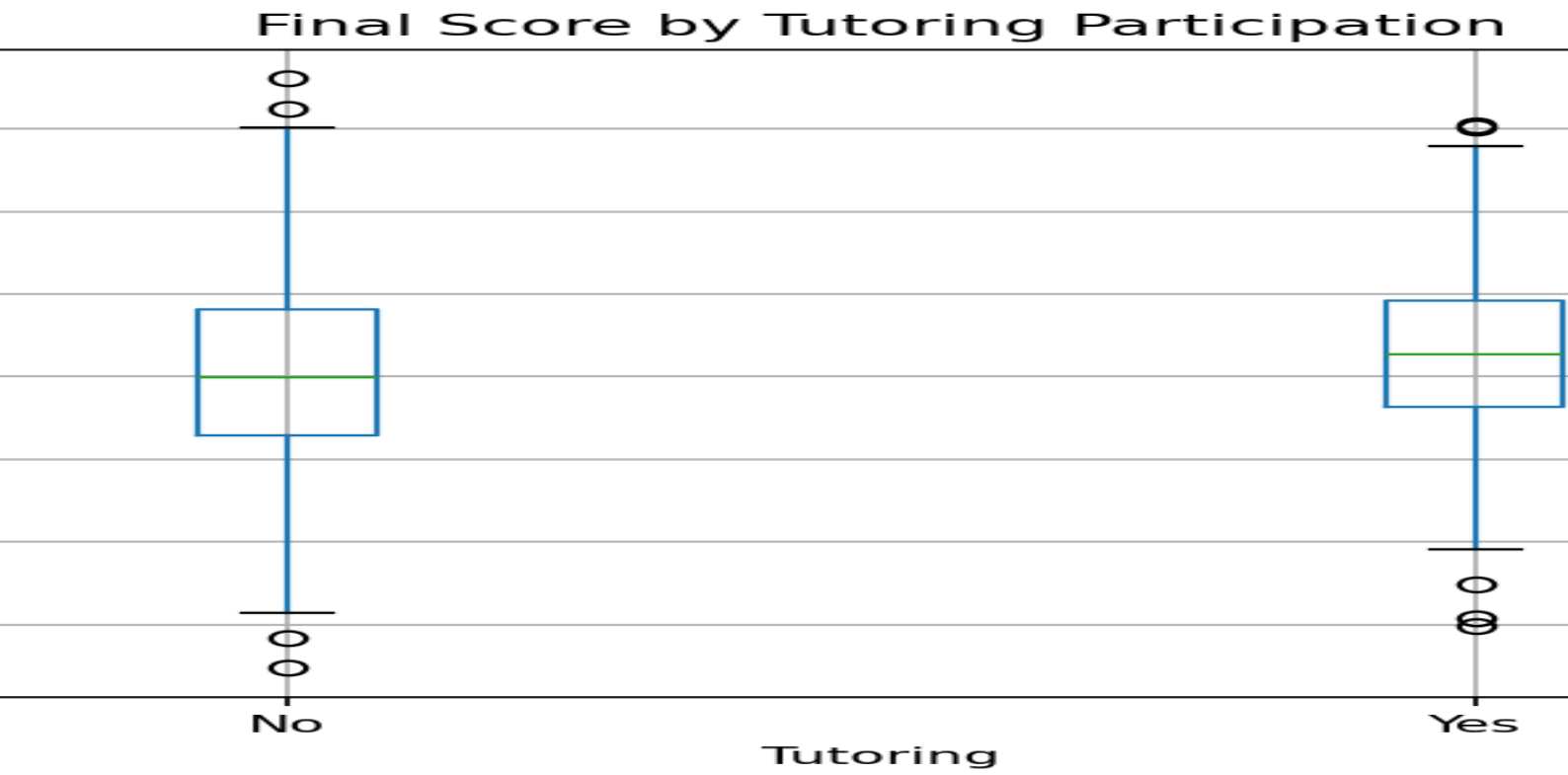


Figure 5 — Score by tutoring participation. Compares medians, spread, and potential outliers between the two tutoring groups. It cannot establish a causal tutoring effect.

5. Model Development and Training

The target variable is **final_score**; all other columns are the input features X. The data is split into 80% training and 20% test records using random_state=42. A scikit-learn Pipeline combines preprocessing and a RandomForestRegressor with 250 trees and min_samples_leaf=2. Random Forest is a useful tabular-data baseline because it can represent nonlinear patterns and averages across many trees.

Numerical columns are median-imputed and standardized. Categorical columns are mode-imputed and one-hot encoded. The pipeline is fitted only on the training split; the held-out test split is used for final evaluation.

6. Evaluation Results

Metric	Test result	Interpretation
MAE	6.206	Average absolute error in score points; lower is better.
MSE	61.778	Average squared error; penalizes large errors.
RMSE	7.860	Error in score points; lower is better.
R^2	0.548	Fit relative to predicting the test-set mean; 1 is ideal.

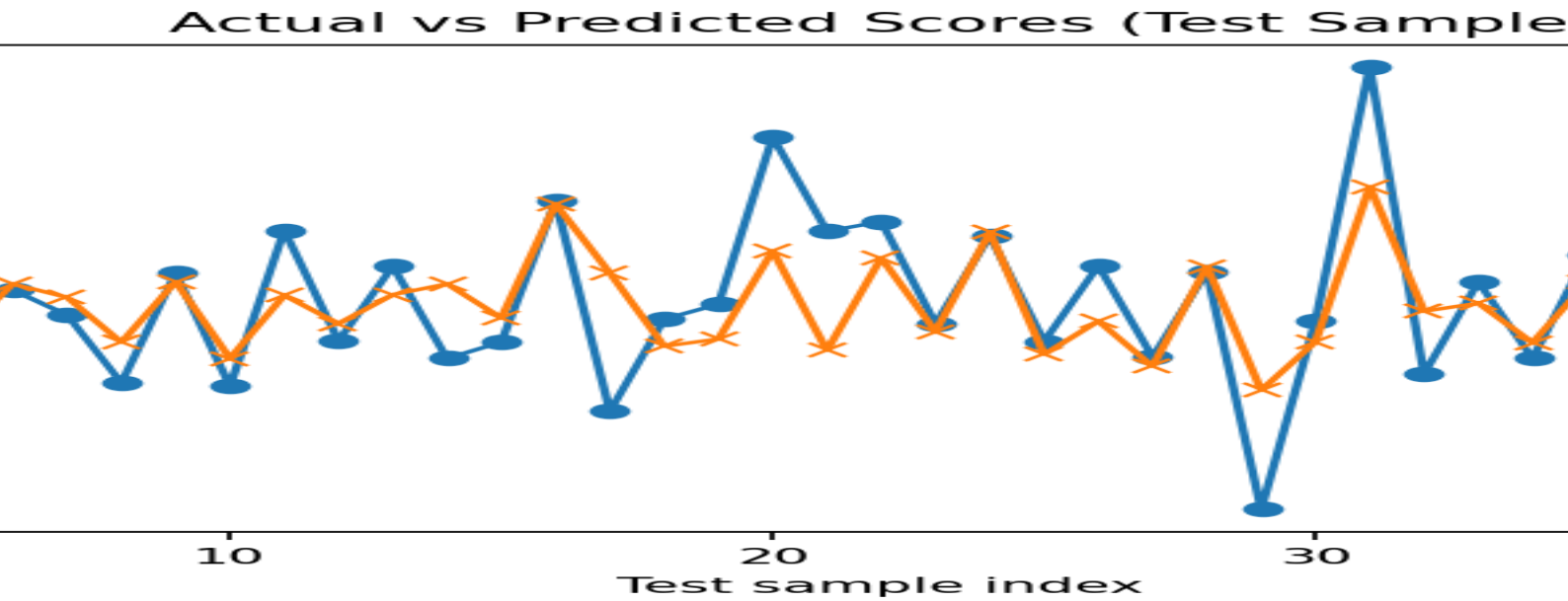


Figure 6 — Actual versus predicted values for a sample of test records. The closer the two lines are, the closer predictions are to observed target values.

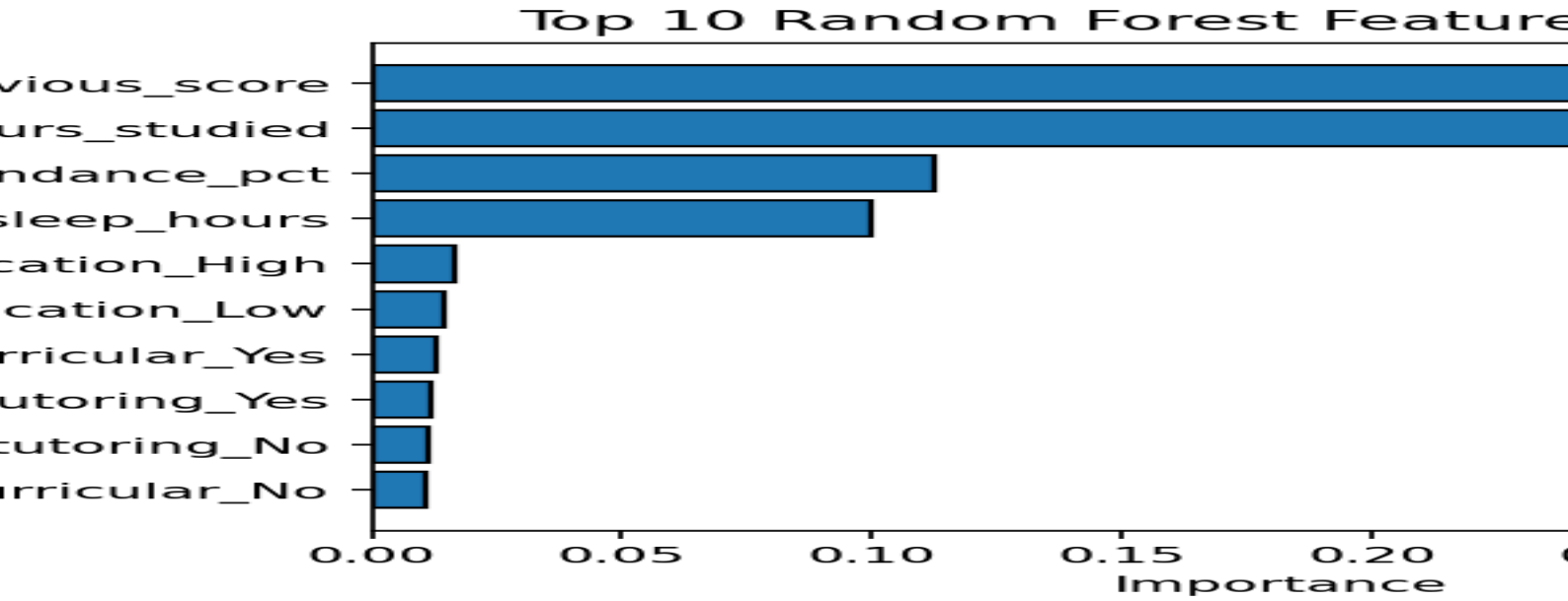


Figure 7 — Top feature importances from the trained Random Forest. Importance is a model-specific measure of predictive contribution, not proof that a feature causes a score change.

The reported metrics are from a single holdout split on synthetic data. They should not be generalized to real students. A stronger evaluation would compare a mean-prediction baseline and alternative models, use cross-validation, inspect residuals, and evaluate performance across relevant groups.

7. Sample Predictions

The notebook and outputs/test_predictions.csv contain test-set actual and predicted scores. Predictions are numerical estimates and should be interpreted as examples of model output, not as official grades or individual educational recommendations.

8. Conclusion

This project demonstrates an end-to-end supervised learning pipeline for a numerical prediction problem. It covers data loading, type and missing-value checks, duplicate removal, imputation, categorical encoding, EDA, an 80/20 split, Random Forest training, predictions, and regression metrics. The accompanying notebook and Python script allow the workflow to be rerun.

9. Future Scope

Future work can use a real, documented, ethically collected dataset; compare Linear Regression, Decision Tree, Random Forest, and Gradient Boosting; tune parameters with cross-validation; inspect residual errors; add confidence intervals; test generalization on a separate cohort; and conduct privacy and fairness assessments. Any real-world use must involve educators, student consent where appropriate, transparency, and human review.

10. Deliverables and How to Run

Files included: dataset (raw and cleaned), Jupyter Notebook, Python script, requirements.txt, README, charts, prediction and metric CSVs, and this PDF report.

Setup: create a virtual environment, install dependencies with `pip install -r requirements.txt`, open Student_Performance_Prediction.ipynb in Jupyter or VS Code, and run all cells from the project root. The script can be run with `python src/student_performance_prediction.py`.

Academic integrity note: Review the code, run it in your environment, inspect the results, and update your name/class details if your institution requires them. Do not claim the synthetic dataset is real.